

HNSS09 · Eye & Ear Histology — Chart

The three layers of the eye and the sensory organs of the inner ear: structure, histology and function | **STEP 1** = beyond class content · **EXAM EMPHASIS** = flagged in class as testable

HoyaDocs
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The Three Layers (Tunics) of the Eye

1 · Fibrous (tunica fibrosa)
sclera + cornea

2 · Vascular (uvea)
choroid + ciliary body + iris

3 · Neural (retina)
neural + non-neural retina

PART	LOCATION	HISTOLOGY (SLIDE ID)	FUNCTION	KEY FACT / CLINICAL
Sclera <i>the white protective coat</i> FIBROUS TUNIC	Posterior 5/6 of the globe; continuous with the cornea at the limbus	Dense connective tissue : flat bundles of type I collagen (some elastic fibers); very few vessels	Protects and shapes the globe; attachment for the extraocular muscles	Limbus (corneoscleral junction) holds the canal of Schlemm + limbal stem cells
Cornea <i>the clear window</i> FIBROUS TUNIC	Anterior 1/6 ; 0.5–1 mm thick	5 layers (see table below). Avascular , transparent, rich in nerve endings . Stroma collagen lamellae at right angles	Main refracting surface of the eye. Endothelium pumps water out → keeps it clear	Avascular → corneal transplants rarely rejected EXAM EMPHASIS
Choroid <i>the blood supply + light sponge</i> UVEA	Posterior uvea, between sclera and retina	2 layers : choriocapillaris (next to Bruch's membrane) + choroidal stroma . Many vessels, melanocytes , fibroblasts, macrophages, lymphocytes	Nourishes the outer retina (O ₂ + nutrients to photoreceptors); pigment absorbs stray light	Blood leaking from the choroid can lift the retina → vision loss. Choroidal melanocytes → ocular (uveal) melanoma . Uveitis = inflammation of iris, ciliary body and/or choroid
Ciliary body <i>the focus + fluid factory</i> UVEA	Between the ora serrata and the iris	Ciliary smooth muscle + ciliary processes lined by a double epithelium : outer pigmented (continuous with RPE) + inner non-pigmented (continuous with neural retina)	Processes secrete aqueous humor and anchor the lens by zonule (oxytalin) fibers . Muscle contracts → zonules relax → lens rounds up → accommodation (near vision)	Aqueous overproduction / blocked outflow → ↑ IOP → glaucoma
Iris <i>the aperture</i> UVEA	Front of the lens; continuous with the ciliary body	Stroma (fibroblasts, pigment cells, vessels) over a double epithelium continuous with the ciliary epithelium. Sphincter = rings near the pupil; dilator = thin longitudinal band on the back	Controls pupil size : sphincter (parasympathetic , CN III) constricts; dilator (sympathetic) dilates. Melanin amount = eye color	Less melanin = blue, more = brown, none = pink (albino). Heterochromia iridis : complete, sectoral or central
Neural retina <i>the light sensor</i> RETINA (INNER)	Back of the eye, from the optic disc to the ora serrata	3 neuron layers : inner ganglion cells (axons → optic nerve) → bipolar cells → outer rods + cones , tips embedded in the pigmented epithelium. Plus horizontal, amacrine, Müller glia (full layers below). Light crosses the whole retina to reach the rods and cones	Phototransduction . Straight path: photoreceptor → bipolar → ganglion ; horizontal + amacrine modulate	Only the neural retina contains sensory cells EXAM EMPHASIS — and within it only rods and cones are photosensitive. Detachment (often trauma) = neural retina splits from the RPE
Non-neural retina <i>the retina past the ora serrata</i> RETINA (INNER)	Anterior to the ora serrata	Double epithelium with no photoreceptors , lining the ciliary body and the back of the iris	Forms the ciliary and iris epithelia	Ora serrata = where the neural retina (photoreceptors) ends and the ciliary body begins

Path of light

Cornea → Anterior chamber → Pupil / posterior chamber → Lens → Vitreous chamber → Retina (ganglion → bipolar → rods & cones)

Anterior (cornea ↔ iris) and posterior (iris ↔ lens) chambers hold **aqueous humor**; the vitreous chamber holds the **vitreous body**.

Cornea, Retina & Eye Landmarks

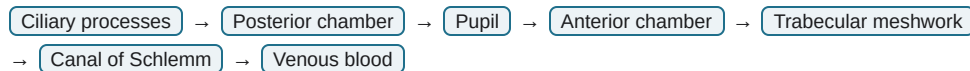
Cornea: five layers

LAYER (OUTSIDE → IN)	HISTOLOGY	KEY FACT
1 • Epithelium	Stratified squamous, non-keratinized ; many free nerve endings	Heals quickly; very sensitive (corneal reflex)
2 • Bowman's membrane	Acellular layer of collagen fibrils	Barrier under the epithelium
3 • Stroma	~90% of thickness; collagen lamellae at right angles , keratocytes	Regular lattice → transparency
4 • Descemet's membrane	Thickest basement membrane in the body	Made by the endothelium
5 • Endothelium	Simple squamous	Pumps fluid out; keeps aqueous out of the stroma → clear cornea

Landmarks

LANDMARK	WHAT IT IS	KEY FACT
Fovea centralis	Pit in the macula lutea ; cones only, avascular ; inner neurons pushed to the foveal rim	Sharpest vision (acuity + color)
Optic disc	Where ganglion axons exit and the central retinal artery enters	No photoreceptors → blind spot
Limbus	Corneoscleral junction	Holds the canal of Schlemm (aqueous drainage)
Ora serrata	Junction of neural retina and ciliary body	Photoreceptors end here
Axes	Anatomic : pupil → optic nerve. Visual : pupil → fovea	Visual axis splits the eye into nasal and temporal halves

Aqueous humor: made → drained



Blocked outflow → ↑ intraocular pressure → optic nerve damage = **glaucoma**. Open-angle: slow, painless, disc cupping. Closed-angle: painful red eye, fixed mid-dilated pupil (emergency) **STEP 1**

Retina: outside (choroid) → inside (vitreous)

LAYER (OUTER / CHOROID SIDE → INNER / VITREOUS SIDE)	CONTAINS	WHY IT MATTERS
Pigment epithelium (RPE)	Melanin-filled cuboidal cells on Bruch's membrane	Absorbs stray light; phagocytoses shed discs (~every 10 days); recycles visual pigment. Bruch's + RPE = blood-retina barrier
Rod + cone segments	Outer segment = stacked discs with pigment; inner = mitochondria, Golgi	Rods: rhodopsin , dim light (100–130 million). Cones: color + acuity (3–6.5 million)
Outer limiting membrane	Junctions between Müller cells and photoreceptors	—
Outer nuclear layer	Photoreceptor nuclei	Detachment splits here from the RPE
Outer plexiform	Photoreceptor → bipolar synapses	—
Inner nuclear layer	Bipolar, horizontal, amacrine, Müller nuclei	Second-order neurons
Inner plexiform	Bipolar → ganglion synapses	—
Ganglion cell layer	Ganglion cell nuclei	Axons → optic nerve
Nerve fiber layer + inner limiting membrane	Ganglion axons heading to the optic disc; Müller end-feet	Superficial bleeds here look flame-shaped

Clinical correlates by layer

CONDITION	LAYER	WHAT GOES WRONG
Glaucoma	Uvea / angle	Blocked canal of Schlemm → aqueous not drained → ↑ IOP → optic nerve damage → blindness
Uveitis	Uvea	Inflammation of iris, ciliary body and/or choroid
Ocular (uveal) melanoma	Choroid	From choroidal melanocytes ; pigmented, can metastasize
Dry AMD	Retina / RPE	Drusen (lipid debris) under the RPE, between retina and choroid; less severe; no specific therapy
Wet AMD	Choroid → retina	Choroidal neovascularization into the macula; leaks, bleeds; more severe; anti-VEGF
Retinal detachment	Retina	Neural retina separates from the RPE (trauma: car accident, fall) → photoreceptors lose support
Central retinal artery occlusion	Retina	Blocked central retinal artery; most often carotid atherosclerosis (embolus)

★ **USMLE high-yield: Fibrous** (sclera + cornea) → **Vascular uvea** (choroid, ciliary body, iris) → **Neural** (retina). **Cornea avascular** → **transplants succeed**. **Descemet's** = thickest basement membrane. **Ciliary processes make aqueous**, which drains via the **canal of Schlemm**. **Bruch's + RPE = blood-retina barrier**; detachment = neural retina off the RPE (floaters, flashes, "curtain"). Fovea = cones only, no vessels. Rods = rhodopsin, night vision; cones = color, center.

Sensory Components of the Inner Ear

Cochlea
organ of Corti → hearing

Utricle + saccule
maculae → linear acceleration, tilt

Semicircular canals
cristae ampullares → rotation

ORGAN	LOCATION	HISTOLOGY	WHAT BENDS THE HAIR CELLS	SENSES	KEY FACT / CLINICAL
Organ of Corti <i>the hearing organ</i> COCHLEA	On the basilar membrane in the cochlear duct (scala media) , filled with endolymph	Hair cells with stereocilia + supporting cells . 4 rows: 1 row inner hair cells + 3 rows outer hair cells. Stereocilia embedded in the overlying tectorial membrane	Sound wave moves the basilar membrane against the tectorial membrane → stereocilia bend	Sound . Inner hair cells transmit (→ spiral ganglion, CN VIII); outer hair cells amplify	Tonotopy : base = high pitch, apex = low . Noise, aging, aminoglycosides kill basal (high-frequency) cells first. Cochlear implant needs a functional auditory nerve EXAM EMPHASIS
Macula <i>the gravity sensor</i> UTRICLE + SACCULE	Walls of the utricle and saccule (vestibule)	Support cells + type I (goblet-shaped, afferent nerve calyx) and type II (columnar, afferent + efferent) hair cells, each with stereocilia + one kinocilium . Covered by a gelatinous otolithic membrane topped with otoconia (CaCO ₃ crystals)	Heavy otoconia lag / slide with gravity → gelatin layer shifts → stereocilia bend	Linear acceleration and head tilt (gravity)	Only the macula has “rocks.” Dislodged otoconia drift into a canal → BPPV (brief vertigo with head position) STEP 1
Crista ampullaris <i>the rotation sensor</i> SEMICIRCULAR CANALS	Ampulla of each of the 3 semicircular canals (at right angles → 3 planes)	Ridge of hair cells + support cells; stereocilia project into a tall gelatinous cupula with no otoliths that spans the ampulla	Head turns → endolymph lags → cupula bends → stereocilia bend	Rotational (angular) acceleration	Macula = Moving in a line; Crista = Circles

Bony vs membranous labyrinth

LABYRINTH	WHAT IT IS	FLUID
Bony	Hollow cavities in the temporal bone : vestibule, semicircular canals, cochlea	Perilymph
Membranous	Epithelial ducts and sacs inside it: utricle, saccule, semicircular ducts, cochlear duct — carry the sensory organs	Endolymph

Endolymph vs perilymph

FLUID	WHERE	IONS
Endolymph	Membranous labyrinth: scala media, utricle, saccule, semicircular ducts	High K⁺ , low Na ⁺ (like ICF); made by the stria vascularis
Perilymph	Bony labyrinth: scala vestibuli + tympani, around the ducts	High Na⁺ , low K ⁺ (like ECF)

Hair-cell signaling & the path of sound

Bend **toward** kinocilium → Tip links pull K⁺ channels open → K⁺ from endolymph flows in → depolarize → Ca²⁺ in → transmitter release → CN VIII fires

Bend **away** → tip links slacken → channels close → **hyperpolarization**. Kinocilium = true cilium; stereocilia = **actin-filled microvilli**.

Stapes → oval window → Scala vestibuli → Scala media / helicotrema → Scala tympani → Round window bulges

★ **USMLE high-yield: Corti = hearing** (tectorial membrane), **macula = linear acceleration/tilt** (otoconia, utricle + saccule), **crista = rotation** (cupula, no otoliths). Endolymph is **high K⁺**; K⁺ influx depolarizes hair cells. **Base = high pitch**, apex = low. Ototoxins: **aminoglycosides, cisplatin, loop diuretics**, high-dose aspirin. **Ménière** = ↑ endolymph (vertigo + tinnitus + SNHL + fullness).

The cochlea in cross-section

SPACE (COCHLEAR CROSS-SECTION)	FLUID	KEY FACT
Scala vestibuli	Perilymph	Starts at the oval window (stapes)
Scala media (cochlear duct)	Endolymph	Holds the organ of Corti ; endolymph made by the stria vascularis on its lateral wall
Scala tympani	Perilymph	Ends at the round window , which bulges out as the stapes pushes in. Joins scala vestibuli at the helicotrema (apex)
Spiral ganglion	—	Cell bodies of the cochlear nerve fibers (CN VIII)